PaperReadyQuestion4

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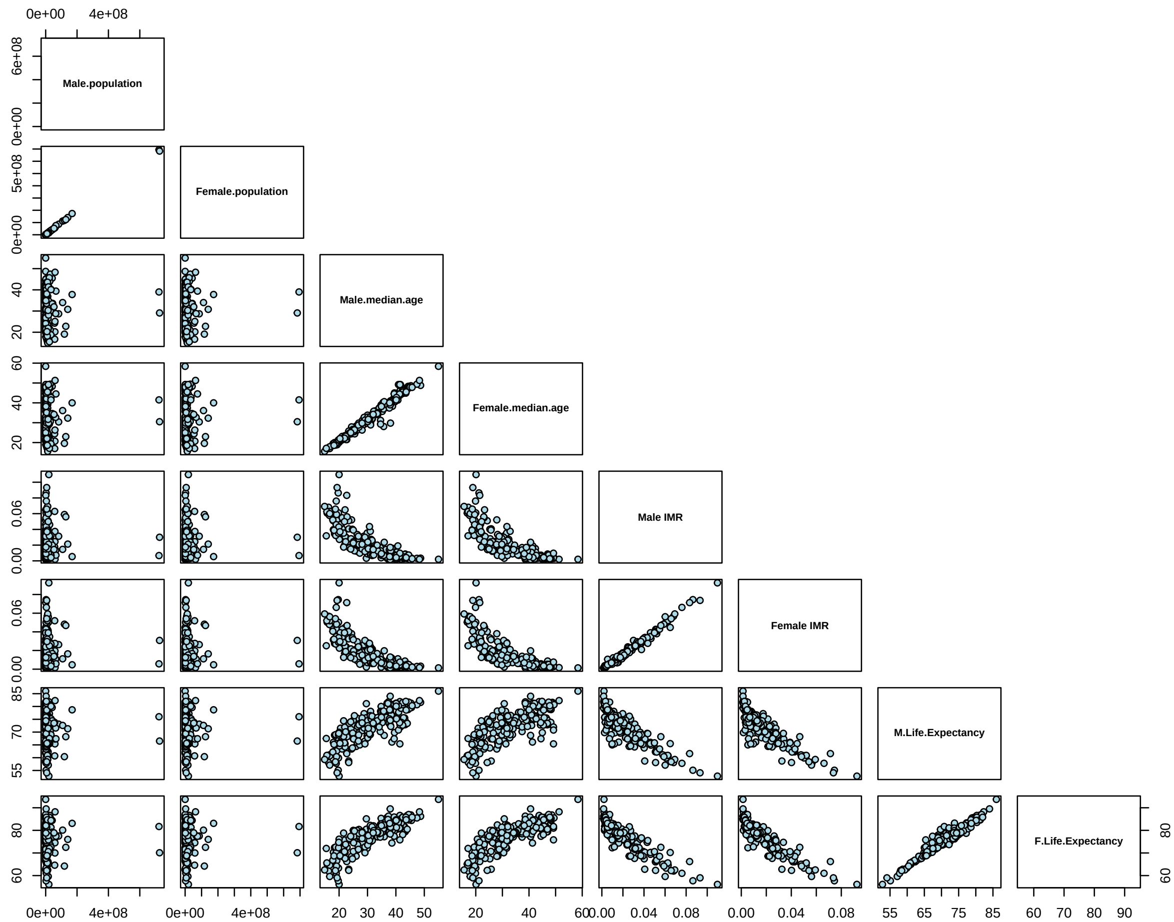
Introduction

In this section we seek to answer the question: "How do selected demographic variables differ between males and females?". The variables we examine are: Population, Life Expectancy at Birth, Infant Mortality Rate, and Median Age. The techniques we use are Pairs Plots, Kernel Density Estimation + Rug Plots, PCA, and Choropleths.

Correlation

```
## [1] "Male.population"                  "Female.population"
## [3] "Median.age.male"                  "Median.age.female"
## [5] "Male.IMR.deaths.per.birth"        "Female.IMR.deaths.per.birth"
## [7] "Male.life.expectancy.at.birth"    "Female.life.expectancy.at.birth"
```

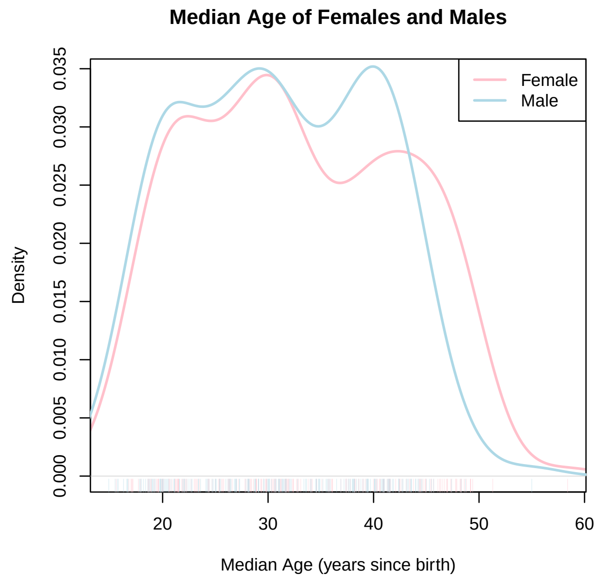
```
## [1] "Male.population"   "Female.population" "Male.median.age"
## [4] "Female.median.age" "Male IMR"          "Female IMR"
## [7] "M.Life.Expectancy" "F.Life.Expectancy"
```

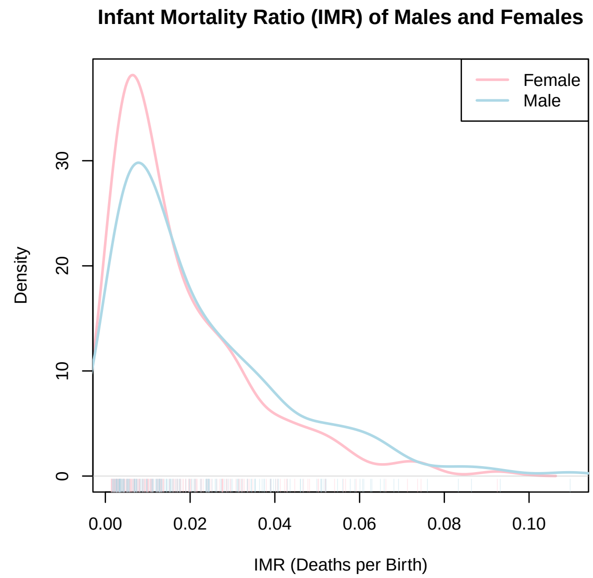


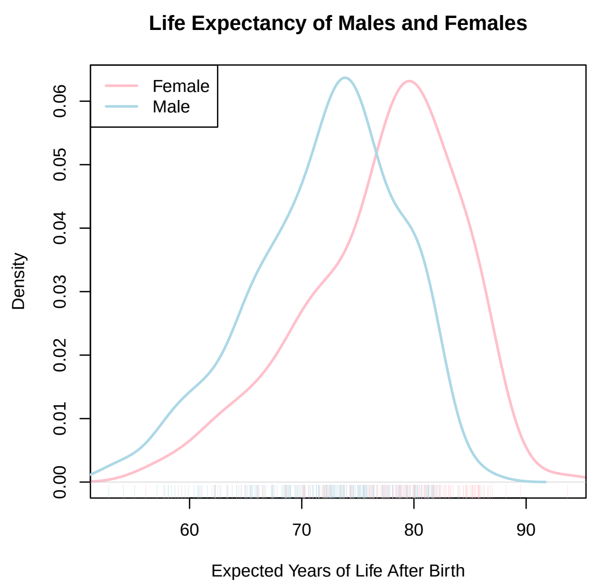
In these plots, one can see a strong, positive linear relationship between the male and female values for each of the variables: population, median age, infant mortality rate, and life expectancy per country. Because of this, we focus on these variables for the remainder of the analysis.

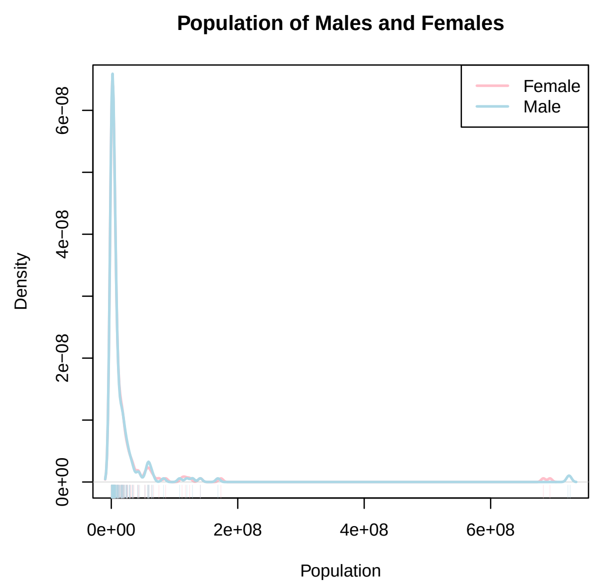
Univariate Statistical Graphics

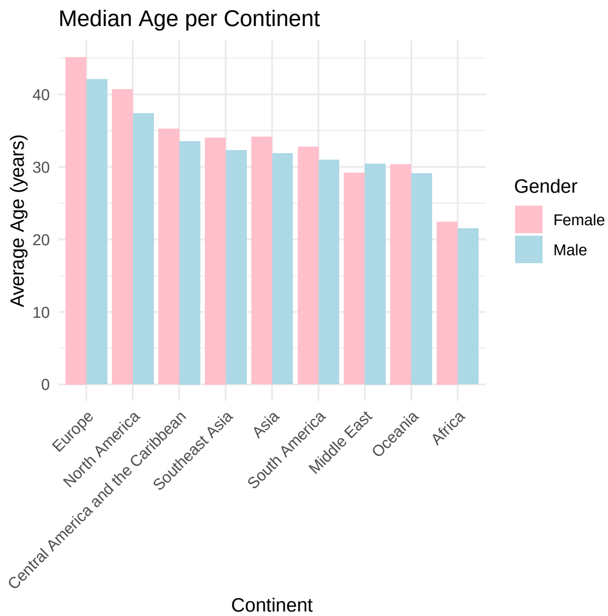
Median Age Across Genders

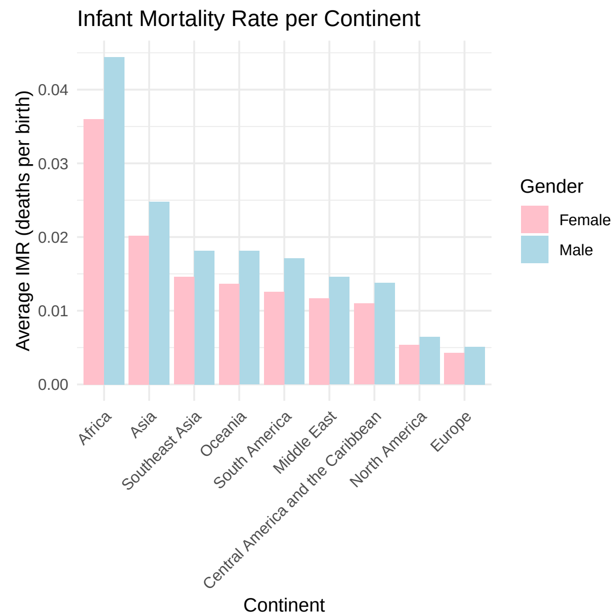


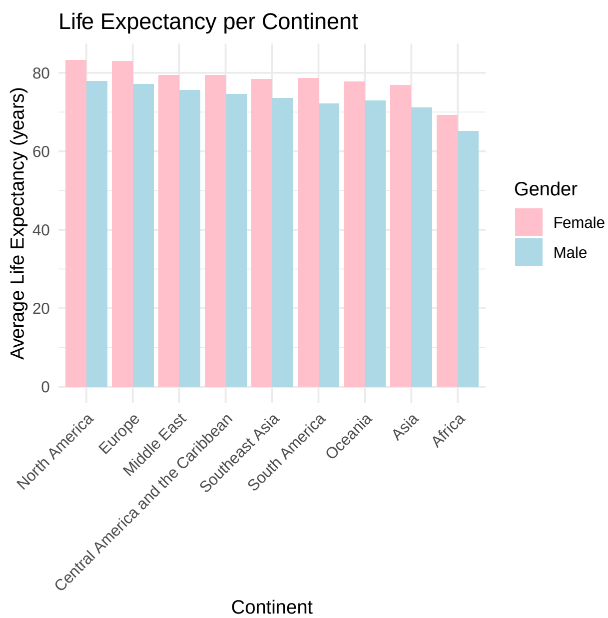


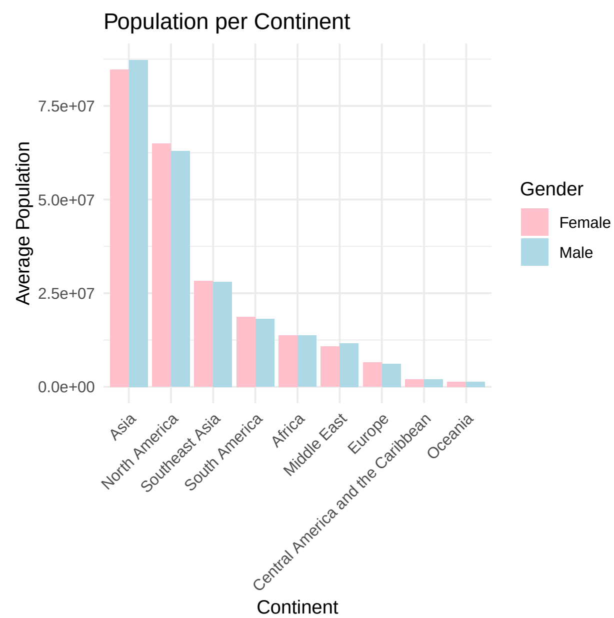












Choropleths

Conclusion